

Technology Stack

Date	01 July 2026
Team ID	SWTID-2026-8998
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap, JavaScript	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, processes user input, and connects the machine learning model with the web application.
3	Database / Data Storage	CSV Dataset	Stores applicant information and historical records for model training and prediction.
4	Cloud / Hosting / Deployment	IBM Watson Machine Learning, IBM Cloud	Deploys the trained machine learning model on the cloud for scalable and real-time credit card approval predictions.
5	Version Control & CI/CD	Git, GitHub	Enables version control, collaboration, and project management during development.

6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Pickle	Used for data preprocessing, feature engineering, model training, evaluation, and saving the best-performing machine learning model.